The four major functions of the cardiovascular system are:

- Transportation of nutrients and gases to every cell and waste products to the lungs and kidneys for elimination.
- Protection of the body from infection and blood loss
- Regulation of body temperature
- Maintenance of fluid and electrolyte balance

Cardiovascular disease (CVD) is a term that is used to refer to a group of disorders of the heart and blood vessels. The most common cardiovascular diseases include atherosclerosis, coronary heart disease (CHD), arrhythmia (irregular heart beat), heart failure, hypertension, and congenital heart disease (heart disease at birth).

The most common reason for heart disease is a build-up of fatty deposits on the inner walls of blood vessels which is called **'atherosclerosis'.** The risk of CVD may be increased by smoking, hypertension, elevated blood lipids, an unhealthy diet, lack of exercise, and obesity. **CVD remains the leading cause of death and disability in our society.**

11.8 Pathophysiology of atherosclerosis

Normal blood vessels are flexible and elastic. **'Arteriosclerosis'** is a condition where the arteries harden, making the passage of blood difficult and sometimes impossible. It occurs when the blood vessels that carry oxygen and nutrients from the heart to the rest of the body become thick and stiff. **'Atherosclerosis'** refers to the accumulation of fats, cholesterol and other substances in the inner lining of arteries over time leading to the formation of a **'plaque'.**

As the plaque increases in size, it gradually reduces the size of the lumen of the artery (space within tubular structures, in this case, artery) and consequently the amount of blood flow through it. The reduced blood flow causes an inadequate supply of nutrients and oxygen to the tissues. This condition is called **'ischemia'.** The reduced oxygen supply causes pain. If the pain occurs in the chest and radiates down the left arm, it is called **'angina pectoris'**.

When the lumen of the coronary artery which is the blood vessel that nourishes the heart muscles become too narrow, it cuts off the blood flow to the heart muscle (myocardium) depriving it of oxygen. This is termed as a **'heart attack or myocardial infarction'.** When blood flow to the brain is blocked in this way, it results in a **'stroke'.**

Pathophysiology of atherosclerosis

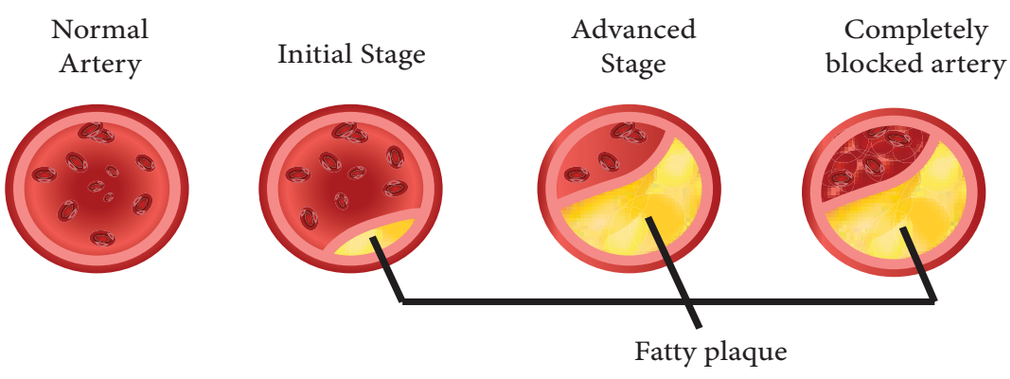


Fig 11.4 Stages of atherosclerosis

UNIT - 11.indd 181 25-02-2019 15:38:21